

Digital Video Restoration by Weighted Fusion of Fast Marching in the Wavelet Domain and Block Translation

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Abstract

In the video restoration, a fast marching method can restore small areas quickly, but there will be significantly blurring when restoring large damaged areas of video frames. To solve these problems, the paper presents a weighted fusion algorithm of fast marching based on the wavelet domain and block translation. Firstly, we can take a wavelet transform to the damaged area of the current frame where a fast marching method will be used to restore the low-frequency part, and the high-frequency part will be obtained from the low frequency part of the reconstruction reconditioned. In view of the temporal correlation of adjacent frames, we can select any one piece as a template before calculating pixel difference between the current frame with damaged area and the full next frame. Then move the full area of next frame to the corresponding position of current frame. The final results obtained by the weighted fusion. The experiments of video restoration verify feasibility, efficiency and Practicality of the proposed algorithm .
Keywords: video restoration, wavelet transform, fast marching method, weighted fusion

1. Introduction

The last several decades have witnessed rapid growth in the popularity of digital images in painting and digital video restoration, and they are the important issues in the field of computer vision. The loss of some information and frames damage appear in the process of video acquisition, transmission and compression. Therefore, we need restore the damaged area to ensure quality of video. Digital video restoration is divided into two: those which eliminate mixed noise and those which restore small area.

In according to restoration algorithm, there are two classes: one is based on a partial differential equation for the small damaged area in video frames, and one is based on texture synthesis [1] for the big damaged area in video frames.

The restoration based on a partial differential equation was proposed by M.Bertalmio initially [2-4], and it means BSCB (Bertalmio-Sapiro-Caselles-Bellester) model. By a gradient vector points calculated on the edge of damaged area to get direction of isoluxline, it is possible to populate this area. Then Chan and Shan [5] extended a total variation (TV) denoising model and curvature-driven diffusion (CDD) model [6-7] that lay the foundation of 3D video restoration total variation algorithm. Fast marching method (FMM) was put forward by the Telea about setting three weighting factor: direction, distance and level set, and the result was obtained by weighted average treatment to the neighborhood reference pixels [8]. As for texture synthesis and more classic model, Criminals' [9] proposed texture synthesis algorithms which search for the best matching block samples. The advantage is not only unnecessary to split image, but also takes effective consideration of the texture and structural information. Brendt presented a new algorithm inspired by putting forward the repaired piece as an area which remain to be fixed [10], which show the better results of linear structure parts, but increase the operation time. In 2012, An algorithm based on Navier-Stokes equation proposed by Shuaijie L [11] has been used widely to consider both the texture and structure information.

Comparing with the static image restoration, it is harder to restore a digital video and more challenging to implement. Since a digital video has information redundancy in time and space, to compensate each other digital video restoration algorithms based on block motion [12] estimation technique between adjacent frames were proposed. For example Ji [13], who proposed a RPCA robust video recovery algorithms to perform block-matching based on block motion estimation. However, traditional block-matching exists defects such as large consuming and low matching accuracy and so on [14]. In addition to the known area, sparse linear combinations of blocks can fill the unknown region along with some local consistency constraints [15]. For example, a robust joint image alignment algorithm based on low rank sparse decomposition (RASL) [17-18] put forward by Peng [16] can effectively remove occlusion and damage, etc [18]. In

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2012, Shan Jiang presented a method about video restoration on account of surfacelet sparse reconstruction [19] without preprocessing. Since the pixel level frames fusion [20-21] is the basic of other fusion algorithm, it received extensive attention in the field of video frames. Early representative fusion algorithms include Intensity-Hue-Saturation (HIS) transformation [22], principal component analysis [23], high-pass filtering algorithm [24], and linear weighting algorithm [25] and so on. These algorithms are simple frame fusion methods without decomposing video frames required. People study wavelet transform in depth, so video frames fusion based on a wavelet transform [26-27] is certainly worthy of further investigation.

Fast marching method restores defective areas point-to-point by means of the weighted averaging in adjacent regions, so it can restore small areas of video frames effectively. Relevance of video frames is utilized to block motion estimation, and we shift the emphasis toward studying these two aspects and put forward a kind of digital video restoration algorithm based on the wavelet domain fast marching method and block translation method of weighed fusion. Firstly, a wavelet transform is applied to the current frame existing in the damaged area, and the bounds have been derived from the low- frequency part and high-frequency part; the former makes use of FMM while the later achieves reconstruction according to the low-frequency portion that have been reconditioned. Due to the existence of temporal correlation between adjacent frames, this paper selects a little piece of the next frame as a template, and the template can calculate pixel differences between the current frame and rear complete frame; therefore, people tend naturally to resort to block a translation method when panning the un- damaged corresponding region of rear frames to the current damaged region. Good results and restoration consequences have been achieved in the following five experiments.

2. Background and Motivation

Wavelet transforms can effectively extract the local characteristics of the signal, and can detect subtle signal changes while some basic signal processing algorithm can't detect the changes. FMM is based on wavelet transform decomposing into low frequency and high frequency with different resolutions. As stated before, taking advantage of FMM developed for low frequency restoration, high frequency can be restored based on the result about a low frequency.

For video frames $f(x, y)$, two-dimensional wavelet transform is defined as [8]:

$$W_j^i f(x, y) = \iint_{R^2} f(x, y) J_j^\theta(x - \hat{x}, y - \beta) dx dy \quad (1)$$

where j is for decomposition scale, and it is on behalf of a low frequency component and three different high frequency components. Wavelet transform is a discrete form. Assuming $h(x)$ is scale function, $j(x)$ is wavelet function, and two-dimensional mother wavelet functions are as follows:

$$\begin{cases} J^0 = h(x)h(y) \\ J^1 = h(x)j(y) \\ J^2 = j(x)h(y) \\ J^3 = j(x)j(y) \end{cases} \quad (2)$$

where their corresponding filter coefficient matrix are H and G ; the original video frame $f(x, y)$ is denoted as A , and then two-dimensional decomposition algorithm is:

$$\begin{cases} A_{j+1} = HA_jH' \\ D_{j+1}^h = GA_jH' \\ D_{j+1}^\theta = HA_jG' \\ D_{j+1}^d = GA_jG' \end{cases} \quad (j=0,1,\dots,J-1) \quad (3)$$

where h, v, d represents horizontal, vertical and diagonal components, respectively. H' and G' represent H and G conjugate transpose matrix [9]. Then we get horizontal, vertical and diagonal components of high frequencies, while most of energy has been concentrated in low frequencies. Below is a brief description of the principle of FMM.

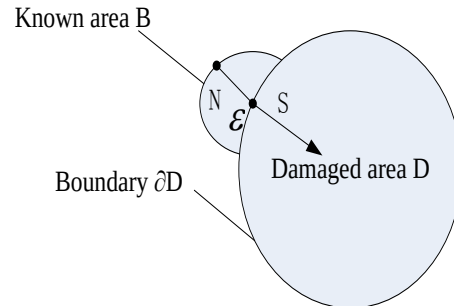


Figure 1: FMM Restoration of the Discrete Pixel

As show in Figure.1, D is a video frame to be restored, ∂D is boundary, S is a pixel to be restored on the boundary, and $B_\epsilon(S)$ is a known set of pixels. T refers to the distance from each pixel of region D to ∂D , and then chooses the path to restore that area following the T values in ascending order, and a shrinking processing of region D . For the sake of simplicity, FMM can be seen as solving the Eikonal equation about the pixel of region D .

$$\begin{cases} |\nabla T| = 1, & \text{Damaged Region} \\ T = 0, & \text{Boundary} \end{cases} \quad (4)$$

D and D have been previously defined, and all of pixels in region D can be presented as three parts:

- 1) KNOWN (known pixels): Pixels are outside the damaged area;
- 2) INSIDE (Unknown pixel): Pixels are in the damaged area, and gray values are unknown;
- 3) BAND (pixels in D): Pixels are in the boundary;

For each pixel S to be repaired, with S in the core, every pixel N as known point has an impact on S, and the following formula interprets the procession:

$$I_N(s) = I(N) + \nabla I(N)(S - N) \quad (5)$$

The weighted function $W(S,N)$ obtained by all known N in neighborhood has the same direction of isolux lines; then by weight average, the value of pixel S has been influenced by N, Formula (6) expresses a whole procedure:

$$I(s) = \frac{\sum_{N \in B_s(s)} W(S,N)(I(N) + \nabla I(N)(S - N))}{\sum_{N \in B_s(s)} W(S,N)} \quad (6)$$

According to the research of telea and others, the weight function $W(S, N)$ is determined by three factors; the direction factor (DIR), distance factor (DST), and the level set factor (Lev)[12]:

$$W(S, N) = \text{dir}(S, N) * \text{dst}(S, N) * \text{lev}(S, N) \quad (7)$$

To sum up, the process of FMM video frame restoration is as follows:

Step 1: initialization. Set the T value of initial boundary and damaged area to 1, and set the T of known area to 0; sort the point called BAND on the stack in ascending.

Step 2: Restoration. Remove the minimum value of T-point on the BAND, and mark KNOWN. Update T value according to "freeze" approach, "light" once will reach a specific pixel within the region to freeze to record this point in time T; followed by rebuild a new cycle boundary, then freeze the recording time T.

Since the video camera's shooting movement, current frame and next frame exist differences to some extent. Block translation prospectively provides video frame information of neighborhood frames in time domain to restore the damaged region in the current frame. Taking current broken frame A and next video frame B for example, the provided two video contents are similar. After grayscale, without considering the precondition of longitudinal movement of the adjacent video frames, choose a part of the frame B as template C, and its upper corner coordinate's corresponding position is a non-broken area in the video frame A. Now suppose the

horizontal position of template C in frame A and B individually to be $(R1, C1)$ and $(R2, C2)$, and calculate coordinate difference $(R0, C0)$. The calculation formula of the coordinate difference is as follows:

$$\begin{cases} R0 = R1 - R2 \\ C0 = C1 - C2 \end{cases} \quad (8)$$

Multiple different coordinates can be obtained according to multiple different templates; as the average value of the multiple coordinate differences is obtained, the relative error of the influence of the coordinate difference to the front and back frames is reduced. Formula (9) describes the process above. Intuitively speaking, the definition is the coordinate difference of three different templates from the current damaged frame A and the next frame B, resulting in three different coordinate differences: (Ra, Ca) , (Rb, Cb) , (Rc, Cc) ; $(R0', C0')$ is the mean of them.

$$\begin{cases} R0' = (Ra + Rb + Rc) / 3 \\ C0' = (Ca + Cb + Cc) / 3 \end{cases} \quad (9)$$

This paper calculates three different coordinate differences for a block translation method. A block translation method makes the restored video frame maintain visual consistency while restoring the true original deleted region as possible. The restoration keeps geometrical consistency of video frames with fast rate of repair process, greatly improving efficiency. Note that selecting template is particularly important, so it will directly affect the quality of block translations. Templates are often difficult to choose when the relative movement is obvious so that a block translation method is mainly used to repair unobvious relative motion of video frames.

Pixel-level fusion has been derived from two different approaches; one makes use of the integration of spatial domain without conversion that belongs to the paper, whereas the other is transform domain fusion. Make some transformation to the frame to be fused as pyramid decomposition transform, or wavelet transform is the premise of corresponding the rule fusion to these decomposition coefficients frame. Then we can inversely transform new video frames from the previous step. Finally, we get a new video frame.

The pixel-level fusion is used to process the corresponding original pixel value from a video frame, by averaging the value from the gradation value of the original video frame multiplied by the corresponding weight coefficients. The following two simple video frames as an example explain the principles. Set the video frame A and B to be fused after gradation processing, their fusion Formula (10) is :

$$\begin{cases} F(i, j) = T * A1(i, j) + (1 - T) * A2(i, j) \\ i = 1 \dots N, j = 1 \dots M \end{cases} \quad (10)$$

where T represents the weighting factor; under normal circumstances the sum of the weighting coefficients $A1$ and $A2$ is 1. Variable (i, j) represents the spatial position of the pixel. $F(i, j)$ is on behalf of the fused image, $A1(i, j)$ is on behalf of video frames by wavelet-based FMM, and $A2(i, j)$ represents video frames by the block translation method restored. It will be an exceptional case when $T=0.5$.

In fact the aforementioned weighted fusion can be seen as a video frame smoothing processing, so it has a strong ability to remove noises as well as high stability to retain more effective frame information as possible. Due to the high sensitivity of the edges, this algorithm has obvious traces of frame splicing for a large difference to participate in the integration of frame. In addition, strict registration should be executed before frame fusions, including translation, rotation angle translation, scaling, etc. Size must be the same in order to get better results in the fusion of video frames. More recently, the combination of weighted fusion and wavelet transform gained prominence in the field of video restoration.

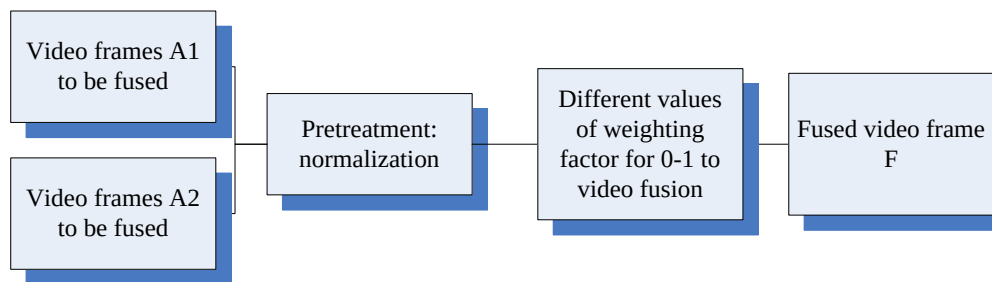


Figure 2: Flow Chart of Weighted Average Fusion

3. FMM Using Wavetet Domain and Block Translation for Weighted Fusion Video Restoration

Digital video restoration is shown in the paper according to Fig. 2. Video frame $A1$ is generated by FMM based on wavelet domain, and Video frame $A2$ comes from a block translation method. Then restoration consequence arrives. Now we should set the weighting factor of $A1$ as T and set the weighting factor of $A2$ as $1-T$. The results shown are encouraging, as they indicate that the weighting factor has taken different values between 0 and 1. Specific steps are as follows:

Step 1: read the video and practice video after framing, and find the frame A to be restored and the next frame B ;

Step 2: Divide videos by wavelet transforms into frames of low frequency and high frequency components, and use a fast marching method to restore the low frequency component of A , which is certainly worthy of further investigation.

Step 3: Take the low-frequency component to horizontal, vertical, diagonal prediction filtering for estimating high-frequency components;

Step 4: Coordinate calculation difference of template C in a restored video frame A and the next frame B ;

Step 5: Translate the part of the full video frame B into the corresponding damaged areas of the video frame A ;

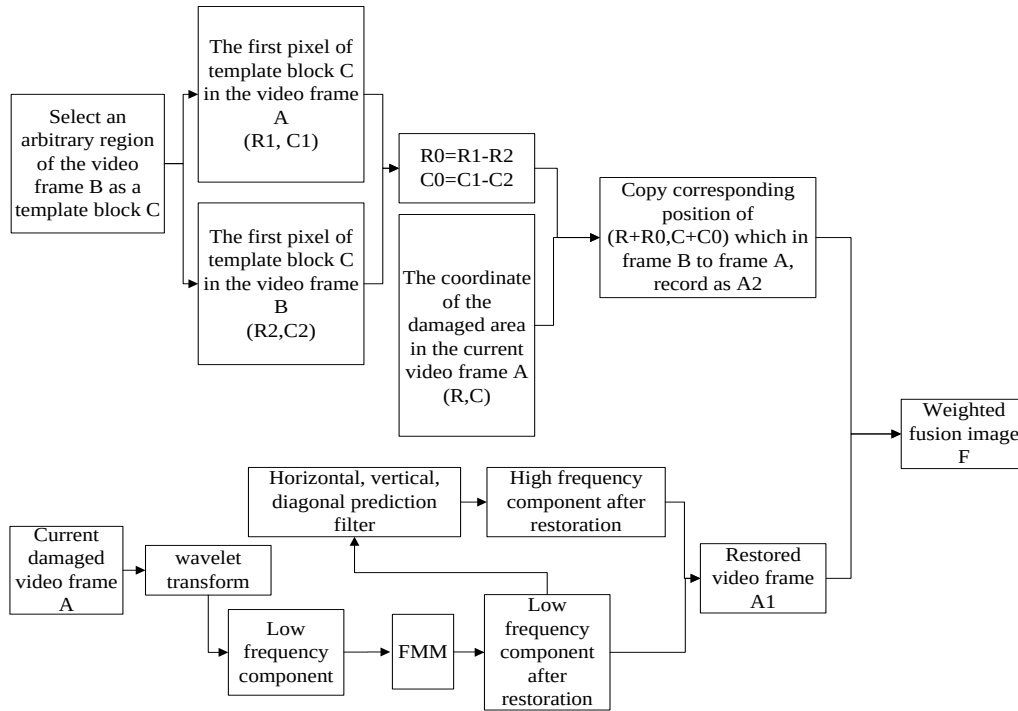


Figure 3: Flow chart of digital video restoration based on FMM in wavelet domain and block translation method

Mean square error (MSE), peak signal to noise ratio (PSNR) and structural similarity (SSIM) are used to evaluate the effect of video frame restoration. The three evaluation indexes are defined as:

$$MSE = \frac{1}{N} \sum_{i=1}^N (A_i - X_i)^2 \quad (11)$$

$$PSNR = 10 \log \left[\frac{x_{\max}^2}{MSE} \right] \quad (12)$$

$$SSIM = \frac{(2\mu_X \mu_A + c_1)(2\sigma_{X,A} + c_2)}{(\mu_X^2 + \mu_A^2 + c_1)(\sigma_X^2 + \sigma_A^2 + c_2)} \quad (13)$$

$$CORR = \frac{\text{cov}(x, y)}{\sqrt{D(x)}\sqrt{D(y)}} \quad (14)$$

A_i can be seen as a fused video frame, X_i is the original video frame, and N represents the pixel points in the restoration area about Formula (11); in Formula (12), $X_{\max}$ is usually defined as 255; while in Formula (13), μ_X is the average of X , μ_A is the average of A , σ_X^2 is the variances of X , σ_A^2 is the variances of A , and $\sigma_{X,A}$ is the standard deviation. $C_1 = (k_1 L)^2$ and $C_2 = (k_2 L)^2$ are used to maintain stable constants; Formula (14) contains that $\text{Cov}(X, Y)$ is the covariance of X and Y , $D(x)$, and $D(y)$ is variance of X and Y , respectively.

4. Experimental Results and Analysis

In order to verify the performance of the algorithm, special design is as follows: the experiment is performed to restore the damaged video. Restoration experiments are performed on a personal computer with 1.70GHz CPU, memory 8GB, pre-installed Windows 7, and ultimate 64 bit operating systems, and the software programming environment is 7.10.0 MATLAB (R2010a).

Experiment one: restore the damaged video frame where the moving object and the background are relatively static, and the camera motion speed is small.

In Figure 4, Figure (a) is the original video frame, the moving object and the background are relatively static, and they overall motion speed to the camera is relatively small; the area of the damaged area is relatively smaller to the video frame area in Figure (b), and the number of missing sample points is 3286; Figure (c) is the video frame which has been restored by the FMM algorithm based on wavelet transforms. Figure (d) shows the video frame is applied by the block translation. Figure (e) is results of weighted fusion by two algorithms. In Table 1, the MSE, PSNR, CORR and SSIM are the commonly used functions to measure the restoration quality of lack digital videos. Using the idea we discussed above, there is not relative motion of video frames due to relatively static of background and moving objects. In other words, R and C are 0. It lays the groundwork for efficient weighted fusions.

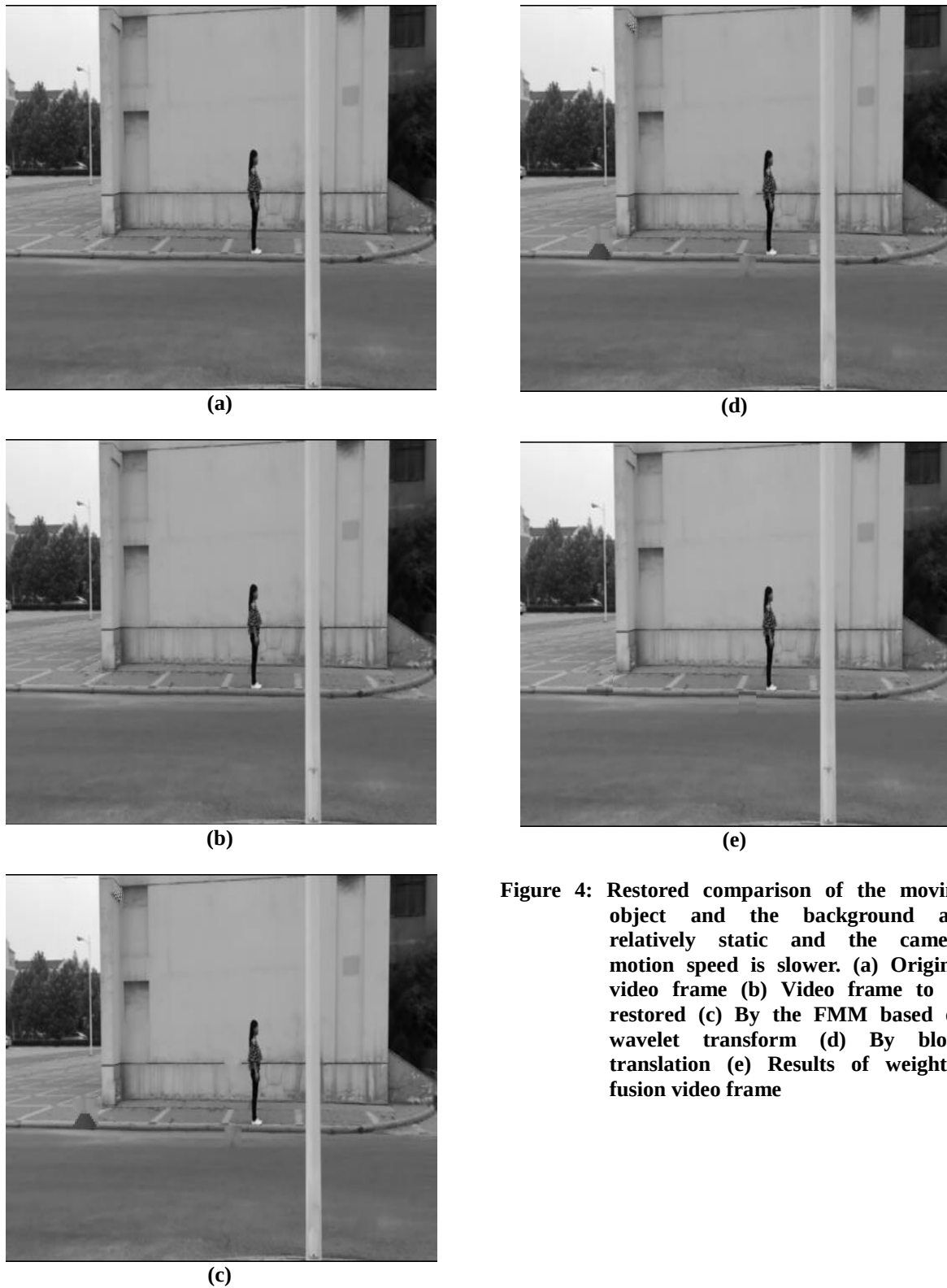


Figure 4: Restored comparison of the moving object and the background are relatively static and the camera motion speed is slower. (a) Original video frame (b) Video frame to be restored (c) By the FMM based on wavelet transform (d) By block translation (e) Results of weighted fusion video frame

Table 1: Performance comparison of algorithm in this paper used in video restoration

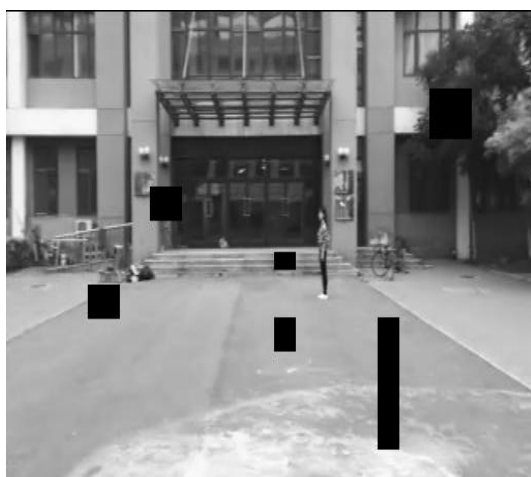
Parameters Algorithm	Before MSE	After MSE	Before PSNR	After PSNR	Before CORR	After CORR	After SSIM
FMM	1.9245e+004	366.1751	5.2877	22.4939	0.9774	0.9785	0.9993
Block translate	1.9245e+004	172.9388	5.2877	25.7519	0.9774	0.9996	0.9988
Weight fusion	1.9245e+004	160.6269	5.2877	26.0726	0.9774	0.9997	0.9994

Experiment two: restore the damaged video frame where the moving object and the background are relatively static, and the camera motion speed is faster than experiment one.

In Figure 5, Figure (a) is the original video frame, the moving object and the background are relatively static, and their overall motion speeds to the camera is relatively fast; the area of the damaged area is relatively modest to the video frame area in Figure (b), and the number of missing sample points is 4946; Figure (c) is the video frame which has been restored by the FMM algorithm based on wavelet transforms. Figure (d) shows the video frame is applied by block translation. Figure (e) is results of the weighted fusion by two algorithms. In Table 2, the MSE, PSNR, CORR and SSIM are the commonly used functions to measure the restoration quality of lack digital videos. According to the results of restoration and display of the parameters, and from the block translation method, the camera motion speed is faster, which makes the calculation error of the front and rear video frames relatively small, thus obtaining the better restored effect.



(a)



(b)



(c)



(d)



(e)

Figure 5: Restored comparison of the moving object and the background are relatively static and the camera motion speed is faster than experiment one. (a) Original video frame (b) Video frame to be restored (c) By the FMM based on wavelet transform (d) By block translation (e) Results of weighted fusion video frame

Table 2: Performance comparison of algorithm in this paper used in video restoration

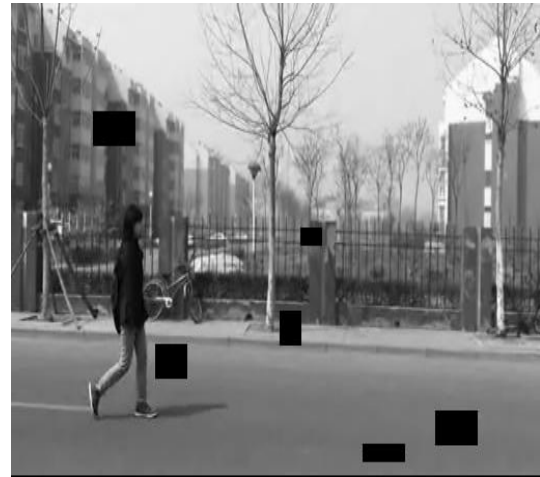
Parameters Algorithm	Before MSE	After MSE	Before PSNR	After PSNR	Before CORR	After CORR	After SSIM
FMM	2.1416e+004	274.0201	4.8235	23.7530	0.9701	0.9705	0.9994
Block translate	2.1416e+004	100.6395	4.8235	28.1031	0.9701	0.9997	0.9993
Weight fusion	2.1416e+004	89.9818	4.8235	28.5893	0.9701	0.9997	0.9993

Experiment three: restore the damaged video frame where the objects move with respect to the background while the camera keeps static.

In Figure 6, Figure (a) is the original video frame, the moving targets keeps moving relative to the background, and the camera is static with jitter. The area of the damaged area is relatively modest to the video frame area in Figure (b), and the number of missing sample points is 3496. Figure (c) is the video frame which has been restored by the FMM algorithm based on wavelet transform. Figure (d) shows the video frame is applied by block translation. Figure (e) is results of weighted fusion by two algorithms. In Table 3, the MSE, PSNR, CORR and SSIM are the commonly used functions to measure the restoration quality of lack digital videos. As previously mentioned, although the camera is still, there is still a jitter, which will be different from the front and back frames so that the pixel difference is applied to the block translation method.



(a)



(b)



(c)



(d)



(e)

Figure 6: Restored comparison of the moving object keep relatively moving to the background and the camera keeps static. (a) Original video frame (b) Video frame to be restored (c) By the FMM based on wavelet transform (d) By block translation (e) Results of weighted fusion video frame

Table 3: Performance comparison of algorithm in this paper used in video restoration

Parameters Algorithm	Before MSE	After MSE	Before PSNR	After PSNR	Before CORR	After CORR	After SSIM
FMM	1.3666e+004	117.3891	6.7743	27.4345	0.9859	0.9862	0.9998
Block translate	1.3666e+004	88.9926	6.7743	28.6373	0.9859	0.9998	0.9993
Weight fusion	1.3666e+004	56.6902	6.7743	30.5957	0.9859	0.9997	0.9994

Experiment four: restore the damaged video frame where the object is moving with respect to the background and the camera keep moving

In Figure 7, Figure (a) is the original video frame, the moving object keeps moving relative to the background with the movement of camera. Facing the complex situation, there will be some certain limitation in restoring the damaged area through block translation. The area of the damaged area is relatively modest to the video frame area in Figure(b), and the number of missing sample points is 5466. Figure(c) is the video frame which has been restored by the FMM algorithm based on wavelet transform. Figure (d) show the video frame is applied by block translation. Figure (e) is results of weighted fusions by two algorithms. In Table 3, the MSE, PSNR, CORR and SSIM are the commonly used functions to measure the restoration quality of lack digital videos. Experiment four has the distinguishing characteristic that exists the relative motion between the moving object and the background with moving cameras. Although the relative error is large from calculating the coordinate difference by the template in the current frame and the next frame, but the overall restoration results are basically satisfactory.



(a)

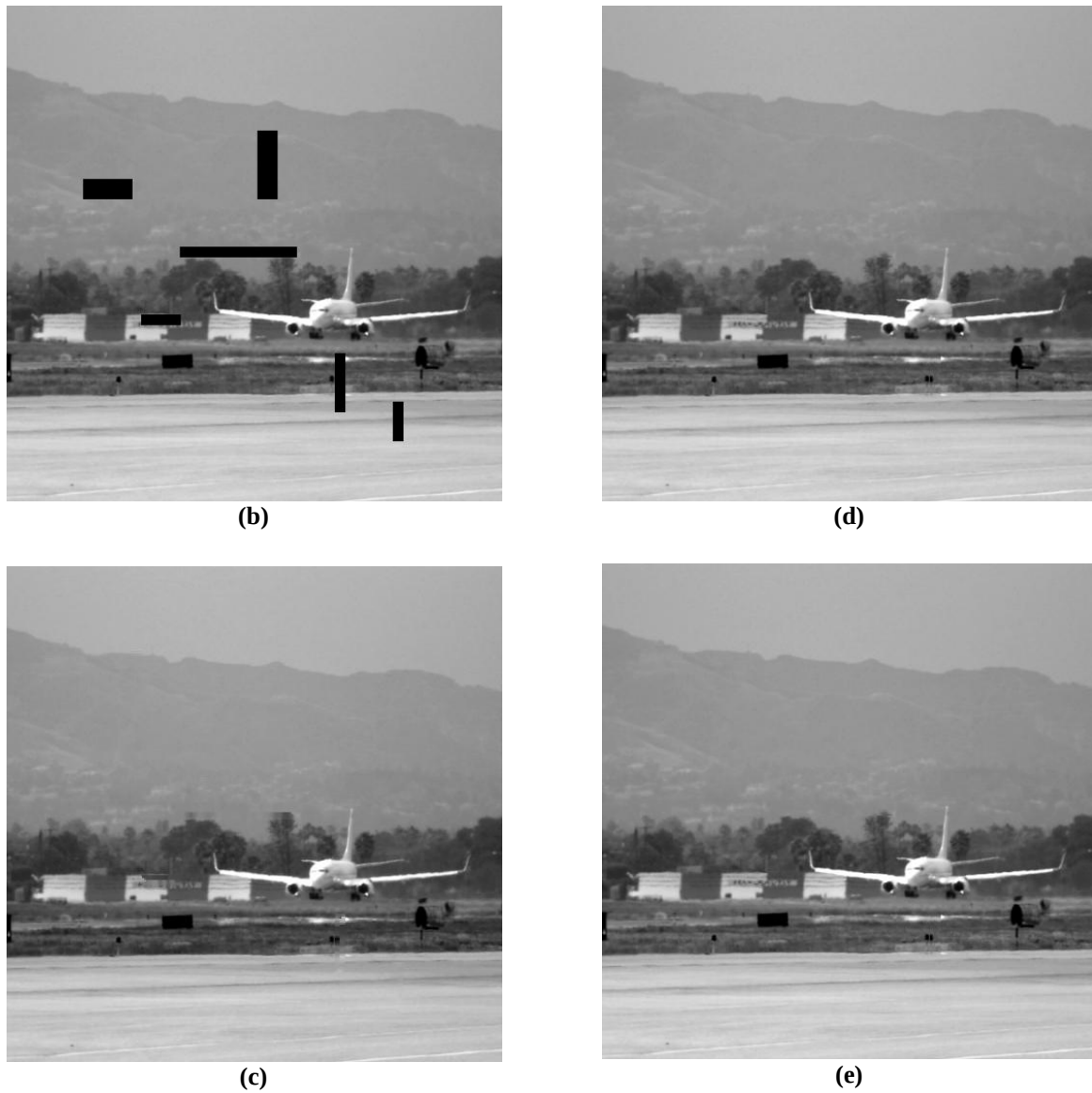


Figure 7: Restored comparison of the moving object keep relatively moving to the background and the camera keep moving. (a) Original video frame (b) Video frame to be restored (c) By the FMM based on wavelet transform (d) By block translation (e) Results of weighted fusion video frame

Table 4: Performance comparison of algorithm in this paper used in video restoration

Parameters	Before	After	Before	After	Before	After	After
Algorithm	MSE	MSE	PSNR	PSNR	CORR	CORR	SSIM
FMM	2.1211e+004	266.6737	4.8653	23.8710	0.9064	0.9094	0.9986
Block translate	2.1211e+004	107.1765	4.8653	27.8298	0.9064	0.9897	0.9891
Weight fusion	2.1211e+004	83.6778	4.8653	28.9047	0.9064	0.9925	0.9891

Experiment five: affect weighting factor T in time correlation.

The process of the fusion algorithm is combination of FMM based on the wavelet transform and block translation algorithm with the essential high efficiency. By exploiting temporal correlation of neighbor video frames, the techniques can efficiently complete the restoration of damaged video frames. However, the weight factor of the fusion process about the two algorithms, to a certain extent, affects the results of PSNR and T , and so on. In this paper, the weight factor of the FMM algorithm is defined as T , while the weight factor of the block translation method is $1-T$. The following Figure 8 shows that the above four sets of experiments in the process of the numerical changes of MSE and PSNR corresponding to the T . Numerical variation between 1 and 0 is found by T . We also observe that the MSE first decreases and then increases, and then decreases with the increase of PSNR according to the increment of T , which is advantageous to prominent optimum points. The analysis of the stables shows that the restored effect is the best when T is selected between 0.3-0.4, and 0.4 is utilized to experiments in this paper.

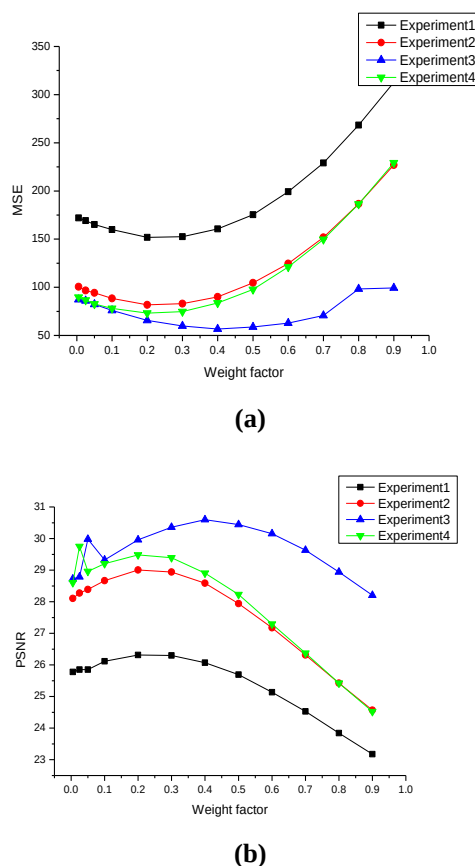


Figure 8: Effected schematic of MSE and PSNR by T
(a) Variation curve of MSE with T transition (b) Variation curve of PSNR with T transition

5. Conclusions

Use the algorithm about the synthesis of FMM based on wavelet transforms and block translation based on temporal correlation. The advantage of this algorithm is to make full use of the temporal correlation of video frames, making up for the FMM blurring on the edges of restoration. The five experiments demonstrate the feasibility and efficiency of our algorithm. While no combination scheme is always better than others in any cases. We intend to further explore this direction of research in how to efficiently restore the moving objects in complex video frames.

Acknowledgments

This work was supported by the China Postdoctoral Science Foundation (2014M561053), and Humanity and Social Science Foundation of Ministry of Education of China (15YJA630108).

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